Autohero Logistics Efficiency Analysis Report

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1. Executive Summary

Autohero, a company focused on car logistics, operates across Germany (DE), Spain (ES), and Poland (PL). This report analyzes key performance indicators (KPIs) over 17 weeks in 2022 to assess logistics efficiency, identify inefficiencies, and provide actionable recommendations. Using SQL queries to compute KPIs and a Tableau dashboard for visualization, the analysis reveals:

- **Key Metrics**: Total bookings of 232,113, workshop deliveries of 2,010, backlog cars of 636,426, and an average backlog age of 1,602 days.
- **Findings**:
  - Bookings peaked in Week 9 (5,684 per country) but dropped significantly in Weeks 14 and 17 (2,478 per country).
  - Workshop deliveries fluctuated, with a low of 13 per country in Week 17.
  - Average lead time improved from 22 days in Week 1 to 1 day in Week 17.
  - The delivery-to-booking ratio remained low, peaking at 0.013 per country in Week 3.
- **Recommendations**: Increase workshop capacity in high-backlog regions (e.g., DE), streamline delivery processes, and monitor KPIs weekly via the Tableau dashboard.

This report provides a clear narrative to help Autohero optimize logistics, reduce costs, and enhance customer satisfaction.

2. Introduction

Autohero aims to streamline its car logistics operations across multiple countries, with a focus on DE, ES, and PL. The company faces challenges such as inconsistent bookings, fluctuating workshop deliveries, and a substantial backlog, which impact operational efficiency and customer experience. This analysis seeks to:

- Track weekly KPIs per retail country.
- Identify trends and inefficiencies in logistics performance.
- Offer data-driven recommendations to improve operations.

The report leverages SQL-derived data and Tableau visualizations to present actionable insights based on 17 weeks of 2022 data

3. Methodology

Data Sources

The analysis uses three primary datasets:

- **car_data**: Contains booking details, including booking_date and retail_country.
- **transport_data**: Tracks delivery information, such as delivery_date and destination_type.
- **daily_grid**: Provides weekly boundaries (first_day_of_week and last_day_of_week) for temporal aggregation.

SQL Queries

Three SQL queries were executed to compute KPIs:

1. **Bookings per Week per Retail Country**:
    - Counts bookings by week and country.
    - Example output: Week 1, DE: 4,410 bookings.
2. **First Workshop Entry with Average Lead Time**:
    - Calculates workshop deliveries and average lead time (days from booking to delivery).
    - Example output: Week 1, DE: 53 deliveries, 22.14 days lead time.
3. **Weekly KPIs**:
    - Computes bookings, workshop deliveries, lead time, backlog count, and backlog age.
    - Example output: Week 1, DE: 4,410 bookings, 53 deliveries, 2,583 backlog cars.

Full queries are provided in queries.sql (see Appendices).

Tableau Dashboard

The "AUTOHERO Logistics Efficiency Dashboard" visualizes the data with:

- Summary metrics (e.g., total bookings, backlog cars).
- Line charts for bookings and deliveries over time.
- A map highlighting DE, ES, and PL.
- Bar charts for lead time and delivery-to-booking ratio.

The dashboard aligns with the SQL outputs and dataset provided.

4. Analysis and Findings

4.1 Summary Metrics

The following metrics summarize Autohero's logistics performance across 17 weeks:

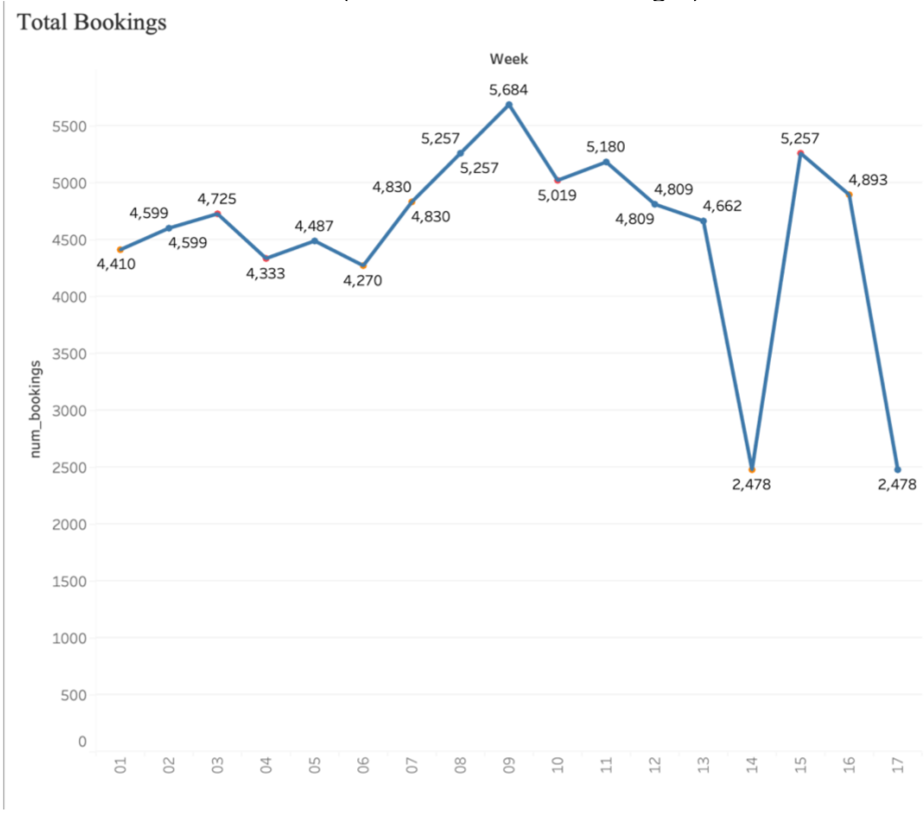
- **Total Bookings**: 232,113 (77,371 per country × 3).
- **Workshop Deliveries**: 2,010 (670 per country × 3).
- **Backlog Cars**: 636,426 (212,142 per country × 3).
- **Average Backlog Age**: 1,602 days (534 per country × 3).

The high backlog and long average backlog age indicate significant delays in delivery, warranting further investigation.

4.2 Trend Analysis

Bookings Over Time

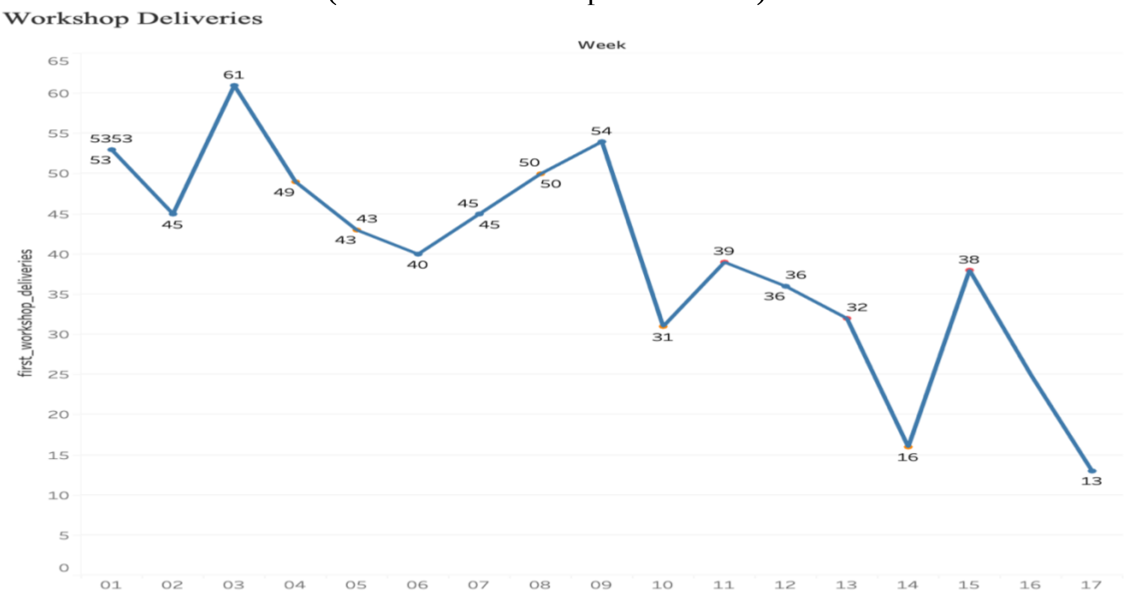
- **Visualization**: Line chart (Tableau: "Total Bookings").



- **Key Observations**:
  - Peak: Week 9 (5,684 per country, total ~17,052 across DE, ES, PL).
  - Lows: Weeks 14 and 17 (2,478 per country, total ~7,434).
  - Trend: Bookings fluctuated, with a noticeable decline in later weeks, possibly due to seasonal factors or capacity constraints.

Workshop Deliveries Over Time

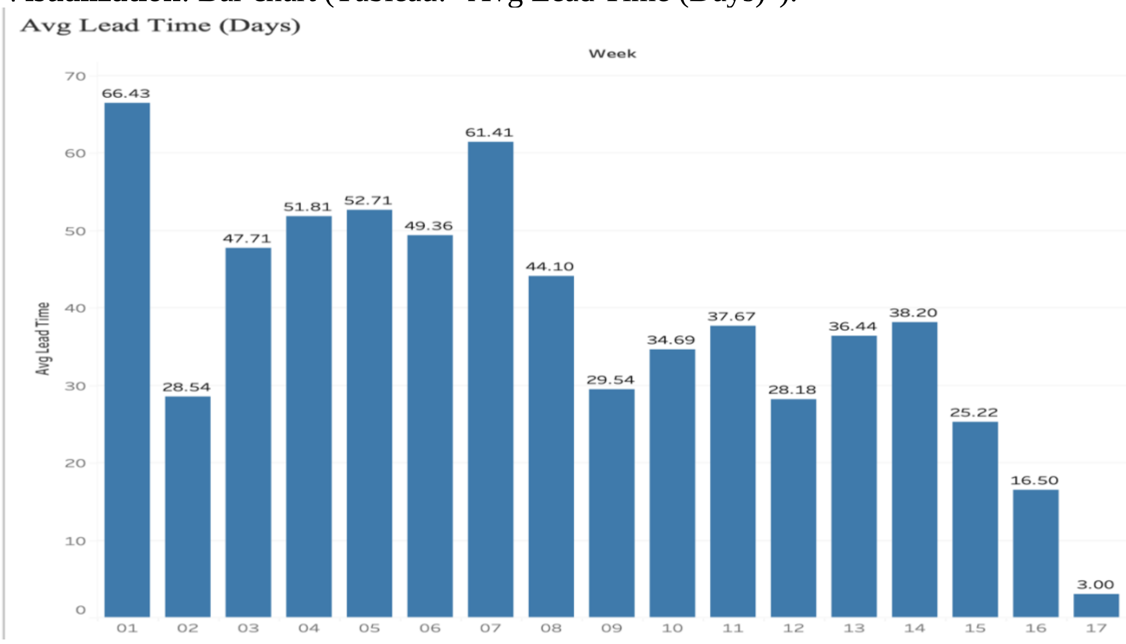
- **Visualization**: Line chart (Tableau: "Workshop Deliveries")



- **Key Observations**:
  - Peak: Week 3 (61 per country, total 183).
  - Low: Week 17 (13 per country, total 39).
  - Trend: Deliveries were inconsistent, with a sharp drop in Week 17, suggesting potential bottlenecks.

Averae Lead Time

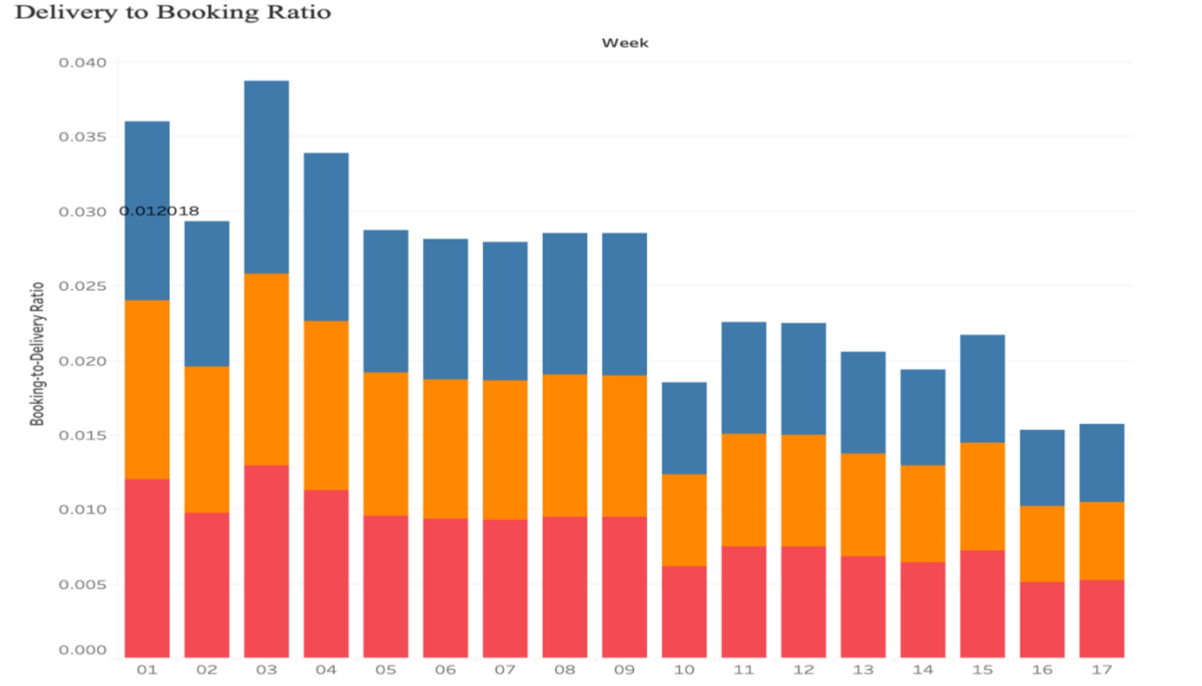
- **Visualization**: Bar chart (Tableau: "Avg Lead Time (Days)").



- **Key Observations**:
  - Week 1: 22.14 days per country.
  - Week 17: 1 day per country.
  - Trend: Lead time decreased significantly, indicating improved processing speed, though early weeks showed delays.

Delivery-to-Booking Ratio

- **Visualization**: Stacked bar chart (Tableau: "Delivery to Booking Ratio")



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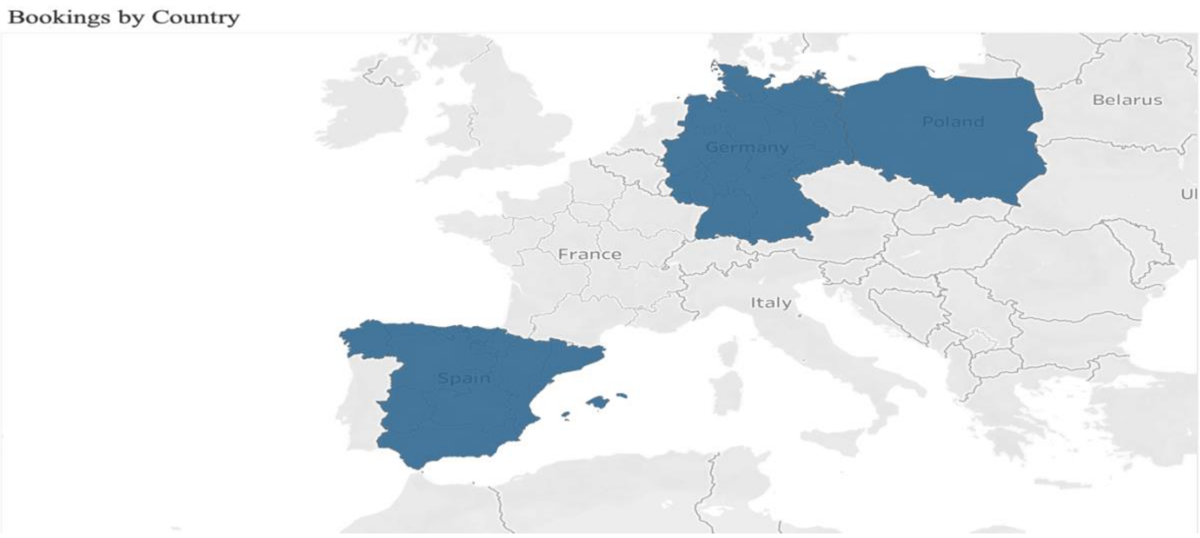
- **Key Observations**:
  - Peak: Week 3 (0.013 per country, total ~0.039).
  - Low: Week 17 (0.005 per country, total ~0.015).
  - Trend: The ratio remained low, reflecting inefficiencies in converting bookings to deliveries.

4.3 Geographical Insights

- **Visualization**: Map (Tableau: "Bookings by Country").



- **Observation**: DE, ES, and PL show similar booking and delivery patterns, but DE has a slightly higher backlog (e.g., 42,455 cars in Week 17 vs. 22,449 in ES and 5,292 in PL), suggesting regional capacity differences.

5. Recommendations

Based on the findings, Autohero can improve logistics efficiency by:

1. **Increasing Workshop Capacity**:
   - Focus on DE, where backlog is highest (e.g., 42,455 cars in Week 17).
   - Expand facilities or staff to reduce backlog and shorten lead times further.
2. **Streamlining Delivery Processes**:
   - Address the low delivery-to-booking ratio (e.g., 0.005 in Week 17) by optimizing workflows and reducing bottlenecks.
3. **Weekly KPI Monitoring**:
   - Use the Tableau dashboard to track trends and respond to issues (e.g., drops in deliveries) in real-time.

6. Conclusion

This analysis highlights inefficiencies in Autohero’s logistics operations, including a significant backlog, fluctuating deliveries, and a low delivery-to-booking ratio. However, improvements in lead time suggest progress in processing efficiency. Implementing the recommendations can help Autohero reduce costs, enhance customer satisfaction, and optimize operations across DE, ES, and PL. The Tableau dashboard serves as a valuable tool for ongoing performance monitoring.

7. Appendices

Appendix A: SQL Queries

- Full queries are available in queries.sql.

Appendix B: Detailed KPI Tables

- See KPI_Overview.xlsx for weekly KPI data by country.

Appendix C: Additional Visualizations

- Additional Tableau screenshots (if applicable) can be included here.